The Natural World · prof.dr. Arjen Siegmann (with help from Claude)

Weitzman's Noah's Ark Problem: Biodiversity, Trade-offs, and the Economics of Conservation

Based on Weitzman (1998), with updated applications and critique

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Purpose. This note introduces Weitzman's (1998) Noah's Ark model—a framework for allocating a conservation budget across endangered species. The model is covered without formal derivations; the focus is on the economic logic, a worked numerical example, and the connection to current conservation practice. Sections 7 and 8 extend beyond the original paper to cover critiques and more recent developments.

1. The Problem

Biodiversity is under severe and accelerating pressure. Current extinction rates are estimated at 100–1,000 times the natural background rate, driven by habitat loss, climate change, invasive species, and over-exploitation (Pimm et al., 1995). Conservation resources—money, personnel, protected land—are finite. Choices must be made.

But choices require criteria. When a conservation agency asks "which species should we prioritise?", the implicit answer in most practice is some combination of threat status (how close to extinction?), charisma (do people care about this animal?), and feasibility (can we actually help?). None of these criteria ask a more fundamental question: *what exactly are we trying to preserve?*

Weitzman's contribution is to answer this question precisely. Biodiversity, he argues, should be understood as *evolutionary distinctiveness*—the genetic and phylogenetic information encoded in the tree of life. Once you accept this definition, a rigorous priority rule follows directly.

2. The Model

2.1 Setup

There is a set $S = \{1, 2, \ldots, n\}$ of species, each with some baseline survival probability $\tilde{p}_i$. A conservation planner has a fixed budget C. Spending c_i on species i raises its survival probability to $p_i > \tilde{p}_i$. The planner's goal is to allocate the budget to maximise *expected biodiversity*.

This raises the central modelling question: what is biodiversity?

Weitzman defines total biodiversity $W(S)$ as the sum of branch lengths on the phylogenetic tree connecting all species in S—the total evolutionary history represented by the surviving set. If species i goes extinct, the loss in diversity is D_i: the branch length that becomes disconnected from the rest of the tree. Expected biodiversity is therefore:

$$\mathbb{E}[W] = \sum_{i \in S} p_i \cdot D_i$$

where p_i is the survival probability of species i. The planner's problem is to allocate the budget C across species so as to maximise this expression. Two things follow immediately: species that are more likely to survive contribute more to $\mathbb{E}[W]$, and—crucially—species with higher D_i contribute more for any given survival probability. This is the core of the model.

2.2 The Library Analogy

Weitzman motivates his diversity measure with an analogy. Imagine a library containing books written in many languages. Some languages are closely related—Spanish and Portuguese share most grammar and vocabulary. Others are more distinct—Finnish and Swahili share almost nothing. If a fire destroys part of the library, what has been lost depends not just on how many books burned, but on *which* ones.

Losing the last surviving text in an otherwise unrepresented language is a greater loss than losing one of a thousand books in a well-documented language. The *informational content* of a book is higher when its language is unique.

Species follow the same logic. The tuatara of New Zealand is the sole surviving member of an entire order of reptiles—its closest relatives disappeared 60 million years ago. Its extinction removes an evolutionary lineage with no substitute. Losing one of several hundred closely related sparrow species is also a real loss, but much of the sparrow's genetic information persists in its relatives.

2.3 Distinctiveness and the Diversity Measure

Weitzman formalises this intuition. Define the *distinctiveness* D_i of species i as the length of the evolutionary branch that would be lost if species i went extinct—in practice, its genetic distance to its nearest surviving relative on the phylogenetic tree.

> *Distinctiveness is what is* lost *when a species disappears, holding everything else constant: the evolutionary branch length that would be removed from the tree of life.*

Figure 1 illustrates the concept. Species B_2 has a very long branch: its nearest relative diverged tens of millions of years ago. Even if B_2 is only moderately endangered, its extinction would erase far more evolutionary history than the loss of A_1 or A_2, which are phylogenetically redundant with each other.

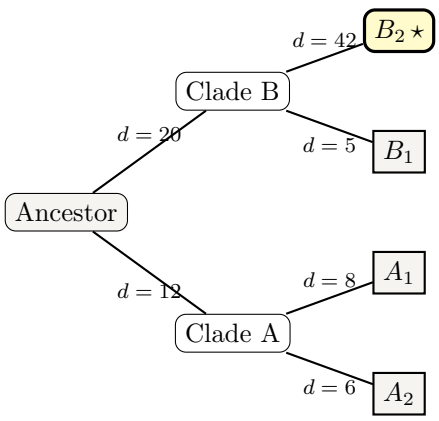

Figure 1: A schematic phylogenetic tree. Branch lengths d represent evolutionary distance (millions of years or genetic divergence). Species B_2 ($\star$) has a very long branch and high distinctiveness $D(B_2) \approx 62$. Species A_1 and A_2 are closely related and phylogenetically redundant: losing either changes total diversity little.

3. The Ranking Criterion

3.1 The priority index

Given limited resources, the planner allocates the budget to maximise expected biodiversity $\mathbb{E}[W]$. Weitzman shows that the optimal solution takes a simple form: rank species by a *priority index* and fund them in order until the budget is exhausted. The index is:

$$\boxed{\Pi_i = \frac{\Delta p_i \cdot (D_i + U_i)}{c_i}} \tag{1}$$

where:

- $\Delta p_i = p_i - \tilde{p}_i$ is the **survivability gain**: how much does spending c_i raise the species' survival probability?
- D_i is **distinctiveness**: the evolutionary branch length lost if the species goes extinct—the species' contribution to aggregate phylogenetic diversity.
- U_i is **direct utility**: the species-specific value to the present generation that is independent of phylogenetic uniqueness—think of a species' role as a keystone predator, its pharmaceutical potential, or the cultural value humans attach to it.
- c_i is the **cost** of the conservation intervention.

This is a *bang-per-buck* criterion: expected diversity gained per dollar spent. Finance students will recognise it as a capital budgeting rule—rank projects by return per unit of scarce resource and fund down the list until the budget binds.

3.2 What the criterion does and does not say

Table 1 summarises how each component enters the ranking.

Table 1: Interpreting the four components of the priority index.

Component	Symbol	High value	Low value
Survivability gain	Δp_i	Conservation is effective	Already safe, or hopeless
Distinctiveness	D_i	Evolutionarily unique	Many close relatives
Direct utility	U_i	Keystone role, high cultural/economic value	Easily substituted
Cost (inverted)	$1/c_i$	Cheap to protect	Requires large investment

Crucially, the criterion does *not* rank simply by extinction probability. A species that is critically endangered—near-certain to die without intervention—could rank below a species that is only moderately threatened, if the endangered species is phylogenetically redundant, prohibitively expensive to protect, or if conservation spending would do little to help.

Note also that D_i and U_i serve distinct purposes. D_i captures what the species contributes to *aggregate* diversity—a substitutable quantity—while U_i captures value that is inherent to the species regardless of how similar it is to others. A species with low phylogenetic distinctiveness can still rank highly if its U_i is large: vultures, for instance, are not especially ancient lineages, yet their functional role as sanitation providers generates very high U_i.

4. A Counterintuitive Result

4.1 Numerical example: the sparrow and the tuatara

Consider two species with the following characteristics:

Species	**Baseline survival** $\tilde{p}_i$	**Gain** Δp_i	**Distinctiveness** D_i	**Direct utility** U_i	**Cost** c_i
Java Sparrow	0.05	0.60	2	0	\$1M
Tuatara	0.40	0.45	80	0	\$2M

The Java Sparrow is far more endangered: only a 5% chance of survival without intervention, versus 40% for the tuatara. Any naïve rule based on threat status alone prioritises the sparrow. Yet the priority indices are:

$$\Pi_{\text{sparrow}} = \frac{0.60 \times (2+0)}{1} = 1.2 \qquad \Pi_{\text{tuatara}} = \frac{0.45 \times (80+0)}{2} = 18.0$$

The tuatara ranks higher—and by a wide margin. It is the sole surviving member of the order *Rhynchocephalia*, a lineage that diverged from other reptiles over 200 million years ago. Its extinction would remove a branch of the tree of life with no close substitute. The sparrow, however endangered, belongs to a species-rich family; its genetic information is largely preserved in its relatives.

> **Key point.** Threat status alone is neither necessary nor sufficient for conservation priority. Evolutionary uniqueness, the effectiveness of conservation spending, and cost all enter the ranking. A moderately threatened species can outrank a critically endangered one.

4.2 When endangered species do deserve priority

The model is not anti-endangered-species—it simply requires discipline. A critically endangered species ranks highly when its survival is sensitive to spending (Δp_i is large), it is phylogenetically distinct, and conservation is relatively affordable. The IUCN Red List is most useful as an input to Δp_i—measuring how much spending can help—rather than as a ranking criterion in itself.

5. Real-World Application: The EDGE Programme

Weitzman's model has been directly operationalised by the **EDGE of Existence** programme at the Zoological Society of London (ZSL), which has maintained ranked conservation priority lists for mammals, birds, amphibians, reptiles, and corals since 2007.

The EDGE score combines two elements:

$$\text{EDGE}_i = \log_2(1 + \text{ED}_i) + \text{GE}_i \tag{2}$$

where ED_i is the *Evolutionarily Distinct* score (essentially Weitzman's distinctiveness D_i) and GE_i is a global endangerment score from the IUCN Red List. The logarithmic transformation prevents the most extreme outliers in distinctiveness from entirely dominating the list.

Selected high-ranking EDGE mammals illustrate the model in action:

Species	IUCN status	Why highly ranked
Long-beaked echidna	Critically Endangered	Monotreme; 50M-year-old lineage
Pygmy hippopotamus	Endangered	Sole living relative: common hippo
Irrawaddy dolphin	Endangered	Sole member of genus *Orcaella*
Solenodon (Cuban)	Endangered	Survived since the Cretaceous

The EDGE list diverges substantially from rankings based on threat alone. Several species rated "Least Concern" appear due to extraordinary distinctiveness; some "Critically Endangered" species rank lower because they belong to species-rich families with abundant close relatives.

> *The EDGE programme operationalises the Weitzman criterion. It is used by conservation NGOs to direct field programmes and funding—a case of economic theory influencing conservation practice directly.*

6. Critique and Limitations

No model survives contact with reality entirely intact. The Noah's Ark framework has attracted substantive critiques on several fronts.

6.1 Is phylogenetic distinctiveness the right objective?

Weitzman equates biodiversity with evolutionary uniqueness. But this is a philosophical choice, not an economic necessity. Alternative frameworks propose:

- **Functional diversity.** What matters ecologically is the diversity of *roles*—pollinator, apex predator, seed disperser, decomposer. A phylogenetically redundant species might fill a unique functional niche, and its loss could trigger ecosystem collapse (Petchey and Gaston, 2002).
- **Option value.** Species may have pharmaceutical, agricultural, or genetic value we cannot currently foresee. Under this view, the value of preservation is an option price—the right, but not obligation, to benefit from a species' properties in the future (Arrow and Fisher, 1974). This framing is familiar to finance students: it treats biodiversity like a portfolio of real options, where irreversibility (extinction is permanent) justifies a precautionary premium.
- **Intrinsic value.** Many conservation biologists and ethicists reject the utilitarian framing entirely. Species have moral standing independent of their usefulness to humans. The model is silent on this, which is either a feature (it stays within economics) or a bug (it misses the most important argument for conservation).

6.2 Survival probability functions are unknown

The model requires $p_i(c_i)$—how survival probability responds to spending. In practice, these functions are deeply uncertain and rarely estimated rigorously. Metrick and Weitzman (1998) found that actual US Endangered Species Act spending was driven more by charisma and political visibility than by any cost-effectiveness calculation. The model tells us what to optimise; it does not tell us how to obtain the data needed to do so.

6.3 The political economy of conservation

Metrick and Weitzman (1998) also showed that conservation spending is heavily skewed toward "charismatic megafauna"—large, attractive mammals and birds—even when smaller, less visible species score higher on objective criteria. The Noah's Ark model is silent on political constraints. A realistic conservation policy must be implementable, which means working with public preferences that are biased toward pandas over beetles, whether or not those preferences are well-founded.

6.4 Ecosystem complementarities

The model treats species as independent: losing one does not change the value of saving another. In reality, ecosystems are deeply interdependent. Losing a keystone species can trigger cascades that affect human welfare in ways that are hard to foresee and harder to value.

A striking recent example is provided by Frank and Sudarshan (2024). In the mid-1990s, the patent on diclofenac—a common anti-inflammatory drug—expired, making cheap generic versions widely available to Indian farmers for treating cattle. Vultures that fed on treated carcasses suffered fatal kidney failure. Within a decade, populations of several vulture species collapsed from tens of millions to a few thousand. The sanitation service the birds had provided—efficiently removing livestock carcasses before they could contaminate water and sustain feral dog and rat populations—disappeared with them. Frank and Sudarshan estimate that the resulting mortality shock caused approximately 100,000 additional human deaths per year between 2000 and 2005, with economic damages of $69 billion annually. India banned diclofenac for veterinary use in 2006, and reintroduction programmes are now under consideration, but recovery will be slow given vultures' low reproductive rates.

The case illustrates two points directly relevant to Weitzman's framework. First, the value of a species can be quantified empirically—this is what the Weitzman criterion requires but rarely gets. Second, the relevant value is not just the species' distinctiveness, but its *functional* role: vultures are not especially phylogenetically distinct, yet their loss was catastrophic. The framework needs to be supplemented with functional ecology to capture cases like this (Faith, 1992).

7. New Insights and Extensions

7.1 Phylogenetic diversity as a portfolio

Recent work in macroecology connects Weitzman's framework to the concept of phylogenetic diversity (PD)—the total branch length of the surviving tree. Maximising PD is equivalent to the Noah's Ark problem under certain assumptions. A phylogenetically diverse set of surviving species is more robust to unknown future shocks—novel pathogens, climate shifts, future agricultural needs—because distinct lineages carry distinct biological solutions.

The analogy to portfolio diversification is direct: in the absence of knowledge about which species will prove most valuable in the future, preserving variance across the tree of life is the robust strategy. The option-value argument for biodiversity conservation has the same structure as the argument for holding a diversified portfolio rather than concentrating in a single asset.

7.2 The 30×30 framework

The 2022 Kunming-Montreal Global Biodiversity Framework set a headline target of protecting 30% of the Earth's land and oceans by 2030. This is a quantity target—area protected—not a quality target. Weitzman's model implies that *where* you protect matters enormously: 30% concentrated in biodiversity hotspots with high phylogenetic distinctiveness is worth vastly more than 30% of randomly selected habitat.

This creates a direct application: given a target of X hectares to protect and a map of species distributions and phylogenetic scores, the Noah's Ark criterion identifies the priority areas. Conservation organisations including the WWF and Birdlife International now use spatially explicit versions of this logic to design protected area networks.

7.3 Rewilding as reinstatement

A growing movement—rewilding—takes a different approach: rather than preventing extinctions one species at a time, restore entire ecosystems and let populations recover. The reintroduction of wolves to Yellowstone in 1995–96 is the canonical example; Ripple et al. (2025) document a large increase in riparian vegetation over the following two decades as elk behaviour changed in response to predation pressure—a trophic cascade propagating through the food web. (The magnitude of the effect remains contested: a comment by MacNulty et al., 2025, challenges some of the quantitative claims, illustrating that even well-studied cases involve significant empirical uncertainty.)

Rewilding is not easily captured by the Weitzman framework, which is fundamentally preventive. The two approaches are complementary: Weitzman identifies what must not be lost; rewilding addresses what can be recovered. In financial terms, one is about limiting downside; the other about rebuilding value. Both have a role in a complete conservation strategy.

8. Summary

The Noah's Ark ranking criterion (equation 1) says: prioritise species by $\Pi_i = \Delta p_i \cdot (D_i + U_i) \,/\, c_i$. Threat status alone is neither necessary nor sufficient. Evolutionary uniqueness, direct species-specific utility, the effectiveness of conservation spending, and cost all enter the priority rule.

Three forces complicate application in practice:

1. We do not know the survival probability functions $p_i(c_i)$ with enough precision to apply the criterion mechanically.
2. Political economy consistently pushes funding toward charismatic species, regardless of their ranking.
3. Ecosystems involve complementarities the species-level model ignores; protecting individual species may be inferior to protecting habitats.

The model's lasting contribution is not a formula to apply mechanically, but a framework for disciplined thinking: it requires specifying what biodiversity *is* and what exactly is lost when a species goes extinct. The EDGE programme demonstrates that the framework can be operationalised.

Discussion Questions

1. A conservation agency has a \$10M budget to allocate between five species. You are given estimates of D_i, U_i, Δp_i, and c_i for each. Apply the Weitzman criterion to rank them. The most endangered species ranks fourth. How would you defend this outcome to the public and to a donor who wants to "save the most threatened animal on earth"?

2. The EDGE programme publishes its ranking list publicly. A wealthy donor insists on funding the most endangered mammal on the list, irrespective of its EDGE score. Using the framework, quantify what is lost by honouring this request. Under what conditions would you agree with the donor?

3. "Phylogenetic diversity is just one possible measure of what makes nature valuable. Functional diversity, ecosystem services, and intrinsic value are at least as important." Does this critique invalidate the model, or merely scope it?

4. The 30×30 target protects 30% of land area by 2030. Using Weitzman's logic, propose a criterion for *which* 30% should be selected. What data would you need, and where would the hardest estimation challenges lie?

5. Is biodiversity conservation more like a capital budgeting problem (allocate a fixed budget across known projects) or a real options problem (preserve flexibility under uncertainty)? What are the implications of each framing for how aggressively we should act *now* versus waiting for better information?

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